

RESEARCH PROFILE

Integrated photonic circuits; optomechanics; microwave-to-optical transduction; nonlinear and quantum photonics; surface acoustic waves; and photonic and mechanical bound states in the continuum.

EXPERIENCE

- Dec 2025–Present** **Postdoctoral Researcher**, Lawrence Berkeley National Laboratory
Berkeley, California
- Jan 2025–Present** **Postdoctoral Researcher**, Department of Electrical Engineering and Computer Sciences, University of California, Berkeley
Berkeley, California
- Sep 2023–Dec 2024** **Postdoctoral Researcher**, Viterbi School of Engineering, University of Southern California
Los Angeles, California

EDUCATION

- 2017–2023** **Ph.D. in Electronic Engineering**, The Chinese University of Hong Kong
Hong Kong SAR
- 2013–2017** **B.S. in Optical and Electronic Information**, Huazhong University of Science and Technology
Hubei, China

PROFESSIONAL SERVICE

Journal reviewer Optics Express · Journal of Lightwave Technology · Journal of the Optical Society of America B · Communications Engineering

JOURNAL ARTICLES & PREPRINTS

† Co-first author. Yue Yu is shown in bold. Publications are listed in reverse chronological order.

1. T. S. Karnik, X. Ren, C.-H. Lee, B.-H. Wu, M. Chaudhari, C. Cheung, J. Wang, S.-Y. Ma, N. Mehrabad, I. Christen, R. Kopparapu, K. Kwon, **Y. Yu**, S. Vadlamani, K. Kunes, Z. Zhuang, D. Englund, Z. Chen, and M. Yu, "18-dB on-chip vacuum squeezing in an adaptively poled lithium niobate waveguide." arXiv preprint, 2026 **PREPRINT** · [Link](#)
2. X. Ren, R. Kopparapu, T. S. Karnik, C.-H. Lee, K. Kwon, C. Cheung, **Y. Yu**, S.-Y. Ma, B.-H. Wu, R. Yin, L. Zhou, Q. Zhuang, D. Englund, Z. Chen, and M. Yu, "Quantum squeezing in an all-resonant periodically poled lithium niobate microresonator." arXiv:2602.22693, 2026 **PREPRINT** · [Link](#)
3. **Y. Yu**, R. Yin, I. Anderson, Y. Wang, J. Kramer, C.-H. Lee, X. Ren, Z. Chen, M. L. Povinelli, D. Wasserman, R. Lu, and M. Yu, "Uncooled low-noise thin-film optomechanical resonator for thermal sensing on lithium niobate." arXiv:2601.02715, 2026 **PREPRINT** · [Link](#)
4. L. Zhou, K. Xue, A. Fallah, L. Liu, C.-H. Lee, K. Kwon, C. Cheung, Y. Li, **Y. Yu**, Y. Lee, S. Zhao, R. Hamerly, E. Waks, D. Englund, C. Sideris, M. Yu, and Z. Chen, "Quantization-aware photonic homodyne computing for accelerated artificial intelligence and scientific simulation." arXiv preprint, 2025 **PREPRINT** · [Link](#)
5. C.-H. Lee, X. Ren, X. Su, W. Lee, Z. Jiang, **Y. Yu**, H. Zhou, Y. Zuo, S. Ou, R. Kopparapu, A. Heiniger, M. Tur, A. E. Willner, Z. Chen, and M. Yu, "Suspended thin-film lithium niobate modulator for broadband mid-infrared light modulation and frequency comb generation." Nature Communications, accepted **IN PRESS** · [Link](#)
6. **Y. Yu**†, R. Yin†, I. Anderson, Y. Wang, J. Kramer, C.-H. Lee, X. Ren, M. L. Povinelli, D. Wasserman, R. Lu, and M. Yu, "Room-temperature low-noise optomechanical platform on thin-film lithium niobate." Light: Science & Applications, accepted **IN PRESS** · [Link](#)
7. R. Yin†, **Y. Yu**†, C.-H. Lee, I. Christen, Z. Chen, and M. Yu, "Fundamental phase noise in thin-film lithium niobate resonators." Laser & Photonics Reviews, e71681, 2025 · [Link](#)
8. S. Guadagnini, Z. Sakotic, **Y. Yu**, M. Yu, D. Wasserman, and M. L. Povinelli, "Suspended, polarization-dependent, subwavelength-perforated metal absorber for mid-infrared bolometry." Nano Letters 25(42), 15182–15188, 2025 · [Link](#)
9. S. Ou, K. Xue, L. Zhou, C.-H. Lee, A. Sludds, R. Hamerly, K. Zhang, H. Feng, **Y. Yu**, R. Kopparapu, E. Zhong, C. Wang, D. Englund, M. Yu, and Z. Chen, "Hypermultiplexed integrated photonics-based optical tensor processor." Science Advances 11(23), eadu0228, 2025 · [Link](#)

10. X. Ren, C.-H. Lee, K. Xue, S. Ou, **Y. Yu**, Z. Chen, and M. Yu, "Photorefractive and pyroelectric photonic memory and long-term stability in thin-film lithium niobate microresonators." npj Nanophotonics 2, 1, 2025 · [Link](#)
11. **Y. Yu** and X. Sun, "Surface acoustic microwave photonic filters on an etchless lithium niobate integrated platform." Laser & Photonics Reviews 18(8), 2300385, 2024 · [Link](#)
12. **Y. Yu** and X. Sun, "Etchless photonic integrated circuits enabled by bound states in the continuum: tutorial." Journal of the Optical Society of America B 40(11), 2801–2808, 2023 **TUTORIAL** · [Link](#)
13. Z. Yu, H. Gao, Y. Wang, **Y. Yu**, H. K. Tsang, X. Sun, and D. Dai, "Fundamentals and applications of photonic waveguides with bound states in the continuum." Journal of Semiconductors 44(10), 101301, 2023 · [Link](#)
14. X.-J. Liu†, **Y. Yu**†, D. Liu, Q.-L. Cui, X.-Z. Qi, Y. Chen, G.-Y. Qu, L. Song, G.-P. Guo, G.-C. Guo, X. Sun, and X.-F. Ren, "Coupling of photon emitters in monolayer WS₂ with a photonic waveguide based on bound states in the continuum." Nano Letters 23(8), 3209–3216, 2023 · [Link](#)
15. Y. Li, Z. Zhang, Y. Wang, **Y. Yu**, X. Zhou, H. K. Tsang, and X. Sun, "Inverse-designed linear coherent photonic networks for high-resolution spectral reconstruction." ACS Photonics 10(4), 1012–1018, 2023 **COVER ARTICLE** · [Link](#)
16. H. Mao†, Y.-X. Ren†, **Y. Yu**†, Z. Yu, X. Sun, S. Zhang, and K. K. Y. Wong, "Neural optimizer for inverse design of complex-modulated holograms implemented by plasmonic metasurfaces." Advanced Photonics Research 4(1), 2200085, 2023 · [Link](#)
17. **Y. Yu**, X. Xi, and X. Sun, "Observation of mechanical bound states in the continuum in an optomechanical microresonator." Light: Science & Applications 11, 328, 2022 · [Link](#)
18. **Y. Yu**†, Z. Yu†, Z. Zhang†, H. K. Tsang, and X. Sun, "Wavelength-division multiplexing on an etchless lithium niobate integrated platform." ACS Photonics 9(10), 3253–3259, 2022 · [Link](#)
19. F. Ye, **Y. Yu**, X. Xi, and X. Sun, "Second-harmonic generation in etchless lithium niobate nanophotonic waveguides with bound states in the continuum." Laser & Photonics Reviews 16(3), 2100429, 2022 · [Link](#)
20. **Y. Yu**, L. Wang, and X. Sun, "Demonstration of on-chip gigahertz acousto-optic modulation at near-visible wavelengths." Nanophotonics 10(17), 4323–4329, 2021 · [Link](#)
21. **Y. Yu**, Z. Yu, L. Wang, and X. Sun, "Ultralow-loss etchless lithium niobate integrated photonics at near-visible wavelengths." Advanced Optical Materials 9(19), 2100060, 2021 · [Link](#)
22. H. Mao†, Y.-X. Ren†, **Y. Yu**†, Z. Yu, X. Sun, S. Zhang, and K. K. Y. Wong, "Broadband meta-converters for multiple Laguerre–Gaussian modes." Photonics Research 9(9), 1689–1698, 2021 · [Link](#)
23. **Y. Yu**, Z. Yu, and X. Sun, "Nonmetallic broadband visible-light absorbers with polarization and incident-angle insensitivity." IEEE Photonics Journal 12(6), 2200807, 2020 · [Link](#)

CONFERENCE PROCEEDINGS & PRESENTATIONS

† Co-first author. Yue Yu is shown in bold.

1. **Y. Yu**†, R. Yin†, Z. Sakotic, I. Anderson, S. Guadagnini, Y. Wang, I. Christen, R. Lu, M. L. Povinelli, D. Wasserman, and M. Yu, "Monolithic room-temperature mid-infrared detector based on a thin-film lithium niobate optomechanical resonator." CLEO: Science and Innovations, paper STh4C.6, 2026 · [Link](#)
2. T. S. Karnik, X. Ren, C.-H. Lee, M. Chaudhari, I. Christen, K. Kwon, K. Kunes, R. Kopparapu, C. Cheung, J. Wang, **Y. Yu**, R. Yin, B.-H. Wu, S.-Y. Ma, Q. Zhuang, D. Englund, Z. Chen, and M. Yu, "Vacuum squeezing in adaptively poled lithium niobate nanowaveguides." CLEO: Fundamental Science, paper FW3A.4, 2026 **POSTDEADLINE** · [Link](#)
3. R. Yin†, **Y. Yu**†, I. Christen, R. Lu, and M. Yu, "Opto-electro-mechanical oscillator on thin-film lithium niobate." CLEO: Science and Innovations, paper SM1A.2, 2026 · [Link](#)
4. S. Guadagnini, Z. Sakotic, **Y. Yu**, M. Yu, D. Wasserman, and M. L. Povinelli, "Suspended subwavelength-perforated metal absorber for mid-infrared bolometry." CLEO, 2025 · [Link](#)
5. X. Ren, C.-H. Lee, R. Kopparapu, C. Cheung, L. Zhou, **Y. Yu**, R. Yin, Z. Chen, and M. Yu, "Few-cycle pulse generation on the thin-film lithium niobate platform." CLEO, paper SS150_3, 2025 · [Link](#)
6. R. Yin†, **Y. Yu**†, X. Ren, C.-H. Lee, Y. Liang, I. Anderson, J. Kramer, R. Lu, and M. Yu, "Intrinsic frequency noise of thin-film lithium niobate platforms." CLEO, paper SS140_4, 2025 · [Link](#)
7. **Y. Yu**†, R. Yin†, I. Anderson, J. Kramer, C.-H. Lee, X. Ren, C. Cheung, R. Lu, and M. Yu, "Room-temperature optomechanical-resonator-based thermal sensor on thin-film lithium niobate." CLEO, paper SS190_1, 2025 · [Link](#)
8. X. Su, C.-H. Lee, X. Ren, Z. Jiang, H. Zhou, Y. Zuo, S. Ou, R. Kopparapu, **Y. Yu**, A. T. Heiniger, M. Tur, Z. Chen, M. Yu, and A. E. Willner, "Mid-wavelength infrared IM/DD communications using an air-suspended thin-film lithium niobate intensity modulator." Optical Fiber Communication Conference, paper Th3J.1, 2025 · [Link](#)
9. C. Cheung, X. Ren, C.-H. Lee, R. Kopparapu, **Y. Yu**, Z. Chen, and M. Yu, "Lithium niobate chip-based ultrafast optical signal processor." CLEO, paper SM4L.5, 2024 · [Link](#)
10. X. Ren, C.-H. Lee, K. Xue, S. Ou, **Y. Yu**, Z. Chen, and M. Yu, "Long-lived photorefractive and pyroelectric effects in thin-film lithium niobate microresonators." CLEO, paper STu3E.1, 2024 · [Link](#)
11. **Y. Yu** and X. Sun, "Surface acoustic microwave photonic filters on an etchless lithium niobate integrated platform." CLEO, San Jose, California, 2023 · [Link](#)
12. Y. Li, Z. Zhang, Y. Wang, **Y. Yu**, X. Zhou, H. K. Tsang, and X. Sun, "Inverse-designed linear coherent photonic networks for high-resolution spectral reconstruction." CLEO, San Jose, California, 2023 · [Link](#)
13. **Y. Yu**, X. Xi, and X. Sun, "Observation of mechanical bound states in the continuum in an optomechanical microresonator." Frontiers in Optics, Rochester, New York, 2022 **POSTDEADLINE** · [Link](#)
14. **Y. Yu**†, Z. Yu†, Z. Zhang†, H. K. Tsang, and X. Sun, "Wavelength-division multiplexing on an etchless lithium niobate integrated platform." Frontiers in Optics, Rochester, New York, 2022 · [Link](#)
15. **Y. Yu**, Z. Yu, L. Wang, and X. Sun, "Ultralow-loss etchless lithium niobate integrated photonics at near-visible wavelengths." CLEO, San Jose, California, 2022 · [Link](#)
16. F. Ye, **Y. Yu**, X. Xi, and X. Sun, "Second-harmonic generation in etchless lithium niobate nanophotonic waveguides with bound states in the continuum." CLEO, San Jose, California, 2022 · [Link](#)
17. **Y. Yu**, Z. Yu, and X. Sun, "Etchless lithium niobate integrated photonics." International Symposium on Lithium Niobate Optoelectronics, Shanghai, China, 2021 **INVITED** · [Link](#)
18. **Y. Yu**, Z. Yu, and X. Sun, "Nonmetallic broadband visible-light absorbers with polarization and incident-angle insensitivity." CLEO, San Jose, California, 2021 · [Link](#)
19. **Y. Yu**, L. Wang, and X. Sun, "Demonstration of on-chip gigahertz acousto-optic modulation at near-visible wavelengths." CLEO, San Jose, California, 2021 · [Link](#)